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Project 5

Introduction

Recognizing handwritten letters is an important problem in optical character recognition, particularly as we move into an age where humanity asks computers to do this task instead. Our project investigates whether the Naive Bayes classification technique for machine learning can distinguish capital letters using numerical descriptions of their shapes. Such recognition abilities would support applications including digitizing forms, processing documents, and making written information easier to search and analyze.

Using the UCI Letter Recognition dataset, we compared Gaussian and Multinomial Naive Bayes classifiers across 26 capital-letter classes. Other research regarding this dataset has shown that the numerical shape features can be used to train classifying models to accurately recognize handwritten capital letters, which initially gave us hope that Naive Bayes would perform well. However, Gaussian Naive Bayes achieved 65.30% accuracy on a grouped test set, while Multinomial Naive Bayes achieved 55.40%. Five-fold grouped cross-validation produced a mean Gaussian Naive Bayes accuracy of 64.34%. The results demonstrate that numerical shape features contain useful information for letter recognition, although some letters remain difficult to distinguish for Naive Bayes showing that it is not the best method for classification with the given features.

Repository: https://github.com/loudpugpug42/cs5830_project5
Slides: Presentation

Dataset

The UCI Letter Recognition dataset contains 20,000 observations. Each observation is described by 16 numerical features representing properties such as character width, height, the number of on-pixels, and horizontal and vertical edge patterns. Each observation is labeled as one of the 26 capital letters, A through Z. The dataset is well balanced between letters, with between 734 and 813 observations per letter, making it suitable for comparing classification performance across classes. All features were recorded as integers between 0 and 15 with no missing values. Because the dataset provides numerical descriptions of each character rather than the original character images, our analysis focuses on how effectively these features distinguish letters.

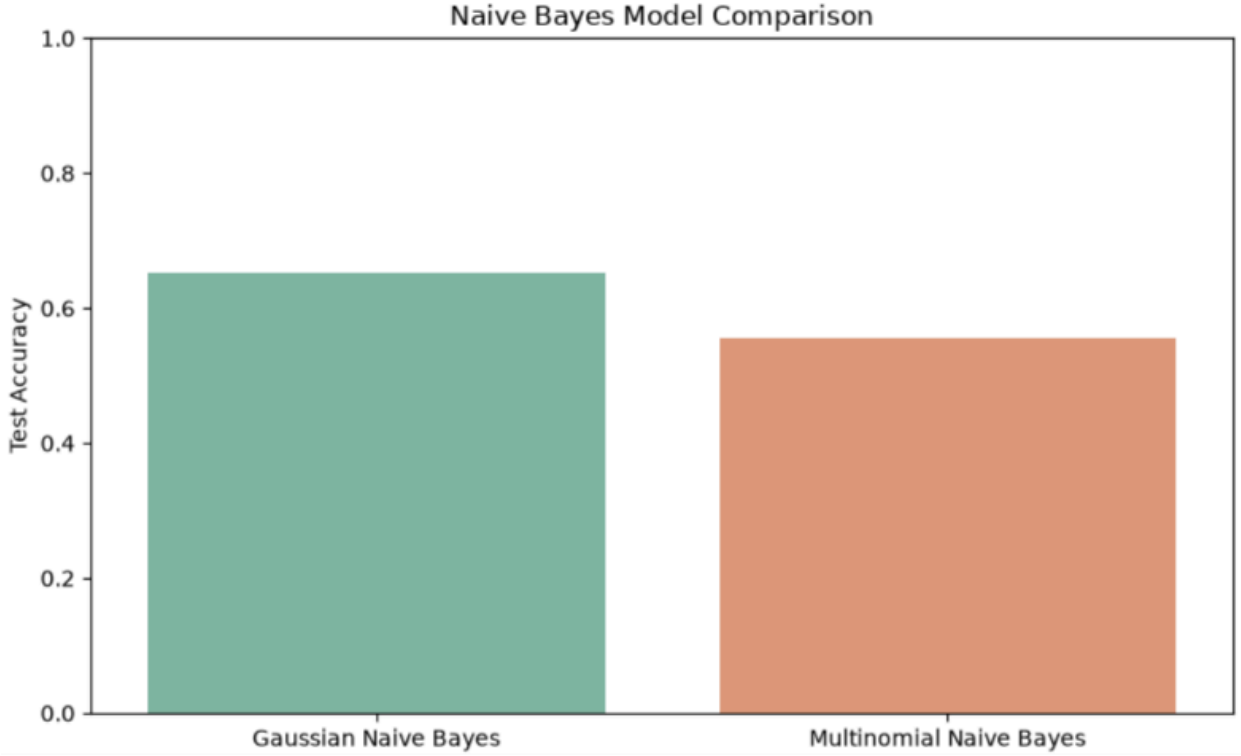
Analysis Technique

We used Naive Bayes because it is a good classification method to estimate the probability of each possible letter given the observed features. Gaussian Naive Bayes models numerical features using class-specific Gaussian distributions which make it a reasonable baseline for numerical shape measurements. Multinomial Naive Bayes was also tested because the features are nonnegative integers which made it mathematically possible, although in our data they represent geometric descriptors rather than event counts. This additional test was performed

because we wanted to observe the difference in capability of these models. Our understanding of the theorems behind these models led us to hypothesize that Gaussian Naive Bayes models would significantly outperform a Multinomial model. Comparing the two approaches allowed us to evaluate this understanding. The models were assessed mostly using accuracy, though we also assessed and recorded precision, recall, F1-score, and their confusion matrices.

Results

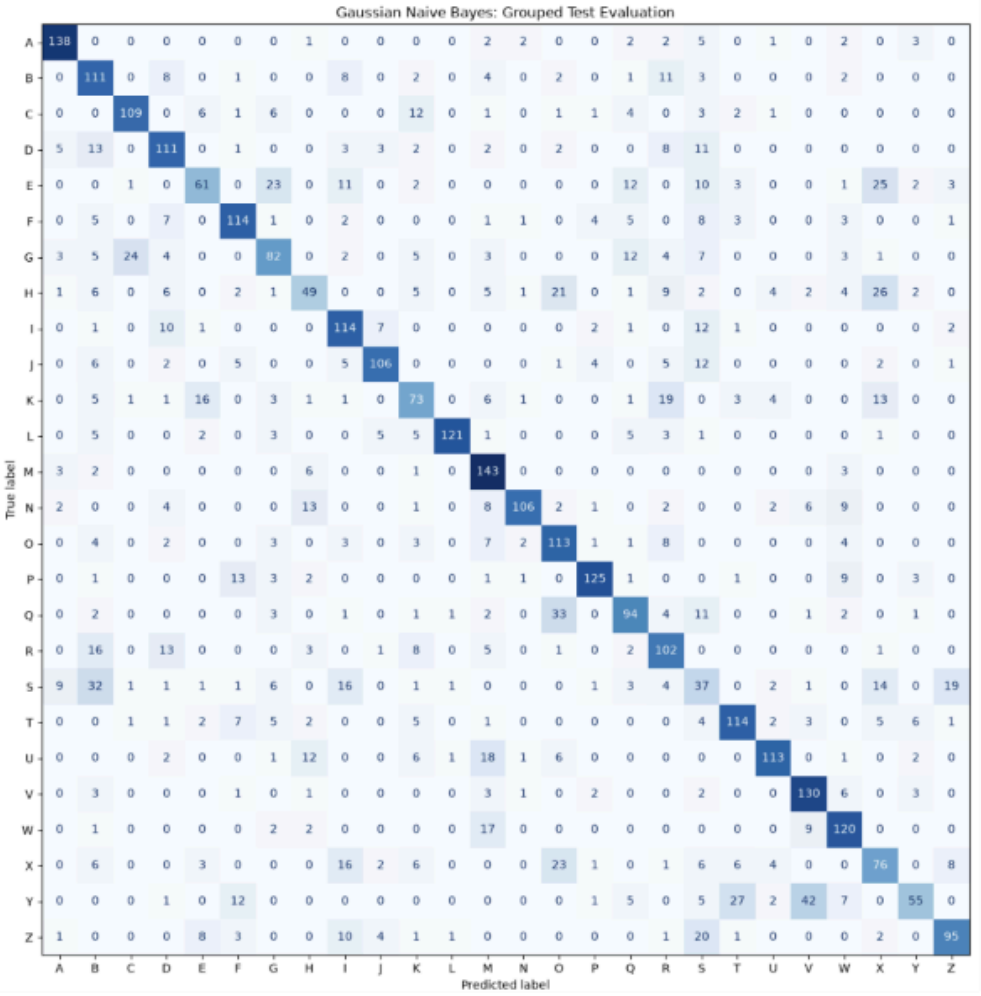
Gaussian Naive Bayes performed better than Multinomial Naive Bayes in both the random and grouped test evaluations. On the grouped test set, Gaussian Naive Bayes achieved 65.30% accuracy, compared with 55.40% for Multinomial Naive Bayes. Five-fold grouped cross-validation gave Gaussian Naive Bayes a mean accuracy of 64.34%, with a standard deviation of 0.59 percentage points. The consistency between the grouped holdout result and the cross-validation results suggests that the model's performance is reasonably stable across different splits. These results support using Gaussian Naive Bayes as the preferred model of the two evaluated approaches, but the low accuracy and relatively minor distance between the performance of each approach lead us to believe that Gaussian Naive Bayes classification will not help a machine differentiate letters in the English alphabet.



Performance between letters was also measured and varied substantially. In the grouped test evaluation, the model achieved F1-scores of 0.87 for L and 0.86 for A, but only 0.24 for S and 0.41 for H. The confusion matrix helped reveal recurring errors, such as Y being classified as V 42 times, Q as O 33 times, and S as B 32 times. These mistakes illustrate how some letters have feature

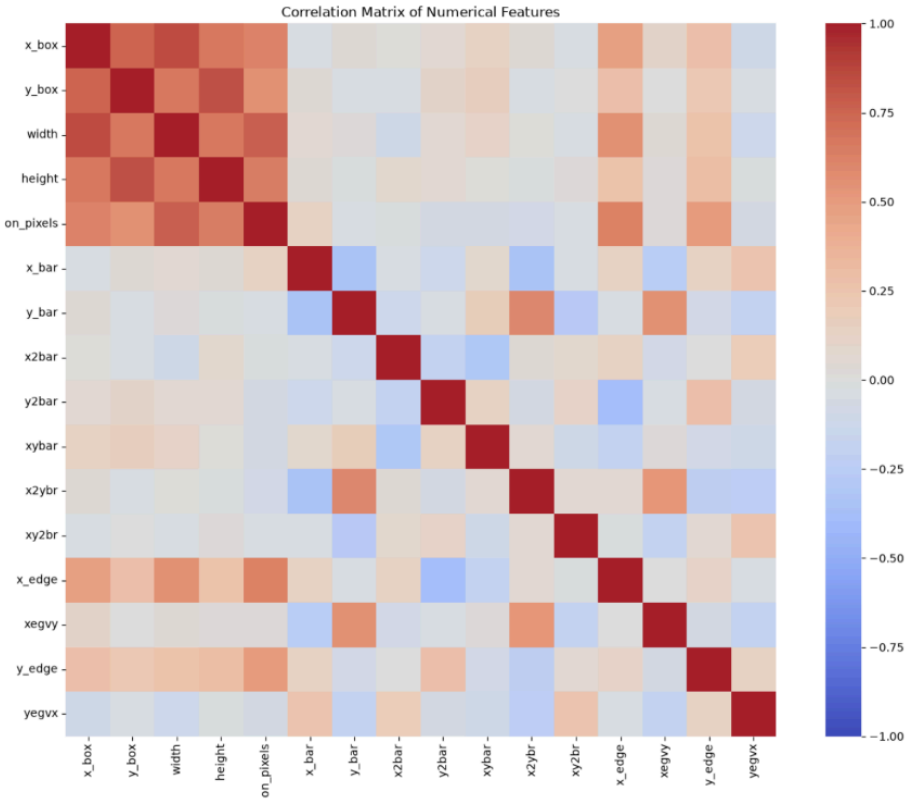
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Most common grouped-test misclassifications:
Y  V    42
Q  O    33
S  B    32
Y  T    27
H  X    26
E  X    25
G  C    24
E  G    23
X  O    23
H  O    21
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patterns that the model can't distinguish. For a practical handwriting-recognition system to be used by stakeholders, including governments attempting to digitize old records or archivals digitizing old documents, the results suggest Naive Bayes predictions would not be reliable enough for such applications that require highly accurate transcription without additional improvements. Other models tested with the dataset such as Random Forest Classification and Xgboost Classification are superior as noted on the page where the dataset is provided.



Technical

The data contained 1,332 duplicate rows, further investigation identified 845 distinct repeated feature vectors. None of these were associated with multiple target labels, so these observations were retained to avoid arbitrarily altering the dataset. Exploratory analysis also identified strong correlations between features, including 0.852 between x_box and width and 0.823 between y_box and height.



An initial stratified 80/20 train/test split was used to train and evaluate Gaussian and Multinomial Naive Bayes. Because identical feature vectors could appear in both subsets under random splitting, we did a second evaluation using StratifiedGroupKFold with feature vectors as groups to prevent identical feature vectors from crossing the training and test boundaries. On the grouped holdout split, Gaussian Naive Bayes achieved 65.30% accuracy and Multinomial Naive Bayes achieved 55.40%. Five-fold grouped cross-validation for Gaussian Naive Bayes yielded a mean accuracy of 64.34% and a standard deviation of 0.59 percentage points. The classification report and confusion matrix were used to examine class-specific performance and identify common mislabeling errors. Strong feature correlations and the Gaussian model's distributional assumptions may limit performance.